

A190 – The Standard Model as Decompactified Projection

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.1 The Thesis

[STIPULATION] The Standard Model is in this theory not a competing framework but the *decompactified projection* of the compact structure.

What in the Standard Model appears as a list of independent inputs – masses, couplings, mixing angles – are in this reading projection images of a structure with fewer degrees of freedom.

.2 What the Projection Explains

[B] Three features of the Standard Model become, in this reading, consequences and not assumptions:

The number three. Three generations follow from the Z_3 structure (A030, A110). In the Standard Model it is an observation without reason.

The hierarchy. The orders of magnitude of the lepton masses follow the ξ -ladder (A100), at the per cent level. In the Standard Model the Yukawa couplings are free parameters without ordering among themselves.

The form of the mass relation. The Koide relation $Q = 2/3$ is an operator condition, not coincidence (A110). In the Standard Model it has no standing.

.3 What the Projection Does Not Explain

[STIPULATION] This section is the longer one, and that is the honest finding.

The gauge group – three convergent arguments.

[B] Argument 1: Charge quantisation from torus flux. The electric charge arises from topologically quantised flux lines through the torus: $\Phi = n \cdot h/e$ with $n \in \mathbb{Z}$. The quantisation is topologically protected – the torus has two non-contractible loops (toroidal and poloidal), the flux through them is invariant under continuous deformations. $U(1)_{EM}$ follows from the torus topology, not from a postulated gauge symmetry.

[B] Argument 2: Colour charge from linking number. The colour charge of QCD arises from the topological linking (linking number) of exactly three flux threads around the torus. Three colours (red, green, blue) = three threads: $SU(3)_C$ is topologically forced by the triple-linking structure of $T^4/\mathbb{Z}_3$.

[S] Argument 3: $SU(27)$ group chain decomposition. From $27 = 3^3$ ($\mathbb{Z}_3$ -torus) follows the chain $SU(27) \supset SU(9)^3 \supset SU(3)^9 \supset SU(3) \times SU(2) \times U(1)$. This makes the SM gauge group the canonical subgroup of the symmetry breaking.

What is still missing: Arguments 1 and 2 deliver $U(1)$ and $SU(3)$ from the torus topology. $SU(2)_L$ follows from the chiral projection (A095). What is not written out: an explicit mapping of the torus topology to the covariant derivatives $D_\mu = \partial_\mu + ig A_\mu^a T^a$ – i.e. the transition from the topological charge structure to the gauge Lagrangian $-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}$.

The coupling constants. Their values and their running are not the subject of the A-series. For α the status is clarified in A130: calibrated, not derived.

The mixing matrices. CKM and PMNS do not appear.

The Higgs sector – consistency check near proof. A complete 1-loop EFT-matching calculation (carried out in the legacy corpus) derives:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}$$

with $\lambda_h = m_h^2/(2v^2)$, $v = 246$ GeV, $m_h = 125.09$ GeV. Numerically: $\xi_{\text{EFT}} = 1.303 \times 10^{-4}$, deviation from $\xi_0 = 4/30000$ only 2.3 %. The $16\pi^3$ structure is the natural result of the 1-loop suppression and NDA normalisation – no fitting.

[S] Two restrictions that together are decisive.

First, the EFT calculation delivers a lepton-specific ξ ($\xi_e \approx 6.7 \times 10^{-14}$, $\xi_\mu \approx 2.9 \times 10^{-9}$), not the universal ξ_0 . The numerically close value $\xi_{\text{EFT}} \approx 1.303 \times 10^{-4}$ is the value for a reference fermion at $\mu = m_h$ in the $\overline{\text{MS}}$ scheme.

Second, the 2.3 % deviation is structurally expected – from two independent reasons that act together:

(a) Unrolled SI values. The formula $\xi_{\text{EFT}} = m_h^2/(64\pi^3 v^2)$ contains $v = 246$ GeV and $m_h = 125.09$ GeV – SI measured values in the $\overline{\text{MS}}$ scheme, scheme-dependent and only at 1-loop level. A 2.3 % deviation corresponds to a 1.1 % deviation in v , which lies within the theoretical uncertainty of 1-loop matchings (typically 1–5 % without NNLO corrections).

(b) Fractal corrections are not cleanly separable in v . The measured VEV $v = 246$ GeV absorbs all SM quantum corrections – the fractal torus structure $D_f = 3 - \xi$ is not separately visible therein but is implicitly contained in the measurement and mixed with the SM loops. The EFT matching in flat space with integer dimension $D = 3$ cannot cleanly extract $K_{\text{frak}} = (1 - 100\xi)$, leading to a systematic inaccuracy. Numerically: $\xi_{\text{EFT}}/K_{\text{frak}}^{3/2} \approx 1.330 \times 10^{-4}$, deviation from ξ_0 only 0.3 % – but that is a post-hoc correction, not a proof.

In short: the deviation is structurally expected, because two different description levels – flat perturbative QFT and fractal torus geometry – do not see the same effective v value.

ξ_{EFT} must therefore not be used to calculate α or E_0 . That would be a circularity trap: one would run SI measured values (v , m_h) through the FFGFT formulas and count the result as confirmation, even though the SI values themselves carry the deviation. The formulas of the A-series use exclusively $\xi_0 = 4/30000$.

ξ_{EFT} is a Wilson coefficient (scheme- and scale-dependent); ξ_0 is the geometric ratio L_0/ℓ_P (scheme-independent). That both agree to 2.3 % – in a calculation using SI measured values with built-in theoretical uncertainty – is the consistency finding. No more, but also no less.

[B] The balance reads: the theory explains the *ordering* of the fermion sector and not the gauge sector. That is a partial statement, and presenting it as an explanation of the Standard Model would be an overreach.

.4 Why the Projection Loses Information

[B] According to A090, the spatial projection (type II) is not reversible. From this it follows that the Standard Model *cannot* fully reproduce the compact structure – regardless of how precisely one measures it.

This is testable, though difficult: what does not completely disappear in the projection leaves a residual of order ξ , and the four prediction quantities of the series are precisely these residuals (A090, A210).

.5 Verification Script

- Numerical verification of projection-chain steps and mass ratios from A090/A100: `python/A_Serie_Skripte/a090_proj.py` (shared with A090).

.6 Open Questions

First, the gauge sector is completely missing. Without it the claim that the Standard Model is a projection is untested for the largest part of the Standard Model.

Second, the projection mapping is not written out. There is in this series no explicit mapping from T^4/Z_3 to the fields of the Standard Model – only the claim that it exists, and examples in the fermion sector.

Third, the statement in its present form is not falsifiable. As long as the mapping is missing, the thesis can be neither confirmed nor refuted. It is a programme.

.7 Supersedes

This document subsumes: Dok. 019 (Standard Model), Dok. 097 (QFT), Dok. 141 (SM parameters), Dok. 298 (SM as projection).

.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.